

Supplementary Information

Cost-Aware Evaluation of Time-Series Foundation Models for Urban Air-Quality and Temperature Forecasting

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This document contains supplementary tables and two supplementary figures referenced from the main text. Table numbers match the main-text citations; the gaps in the numbering (S8, S11) are intentional.

Table S1: City panel. The 29 cities passing the pre-specified quality gate, with country, data-tier (rich/scarce), OpenAQ reference sensor, usable window, and usable hours after gap-gating.

City	Country	Tier	OpenAQ sensor	Window start	Window end	Usable hours
Amsterdam	NL	rich	4235	2025-02-24	2026-07-09	11228
Bangkok	TH	rich	1304255	2023-03-19	2025-04-25	18164
Berlin	DE	rich	1300115	2024-03-23	2026-07-11	19574
Krakow	PL	rich	20854	2018-11-21	2019-12-31	9344
Los Angeles	US	rich	25551	2024-09-12	2025-08-22	8195
Madrid	ES	rich	10378	2025-04-24	2026-01-27	6431
Mexico City	MX	rich	2034	2017-01-30	2017-06-24	3371
New York	US	rich	673	2024-03-07	2025-09-20	13421
Paris	FR	rich	1582598	2023-09-01	2024-03-11	4457
Santiago	CL	rich	1044	2019-01-02	2020-12-11	16288
Seoul	KR	rich	8533919	2024-03-19	2026-07-13	19827
Sydney	AU	rich	21610	2019-12-23	2020-11-08	7277
Toronto	CA	rich	2914	2018-12-16	2020-07-31	13518
Vienna	AT	rich	30223	2025-10-19	2026-07-12	6241
Accra	GH	scarce	30469	2020-05-01	2021-04-24	8160
Addis Ababa	ET	scarce	30190	2016-12-23	2017-05-30	3610
Bogota	CO	scarce	29734	2020-02-10	2020-09-22	5175
Delhi	IN	scarce	35	2016-08-28	2016-12-22	2518
Dhaka	BD	scarce	24434	2023-04-26	2024-05-25	9395
Hanoi	VN	scarce	21632	2019-11-13	2020-07-31	6062
Jakarta	ID	scarce	25072	2019-09-24	2020-07-24	7024
Kampala	UG	scarce	25086	2018-12-16	2019-04-19	2836
Karachi	PK	scarce	23747	2019-11-19	2020-07-01	5214
Lagos	NG	scarce	2251459	2023-04-27	2023-09-15	3378
Lahore	PK	scarce	25135	2024-07-01	2025-02-18	5563
Lima	PE	scarce	4171	2022-03-24	2022-11-06	5227
Manila	PH	scarce	6909373	2024-11-03	2026-02-05	10992
Mumbai	IN	scarce	23320	2020-08-31	2021-01-26	3004
Nairobi	KE	scarce	16091678	2019-08-20	2021-03-10	13576

Table S2: Model configuration per tier. All settings mirror the released harness (`run_forecast.py`) and were fixed before the canonical campaign; none were tuned per city or per domain. The LightGBM `subsample_freq` correction predates the campaign and is logged in the analysis-plan deviations record.

Tier	Setting	Value
Seasonal-naïve	persistence lag	168 h (same hour last week)
LightGBM (specialist)	models	one direct model per lead time (24), retrained per fold
LightGBM (specialist)	n_estimators / learning_rate / num_leaves	250 / 0.05 / 31
LightGBM (specialist)	subsample / subsample_freq	0.8 / 1 (freq added after pre-campaign audit; bagging inert without it)
LightGBM (specialist)	random_state	42
LightGBM (specialist)	features	lagged target, calendar, meteorological covariates
NAS-GRU	architecture	Green-NAS-A: 2 stacked GRU layers, 128 units, direct multi-horizon head
NAS-GRU	optimizer / loss	Adam, learning rate 10^{-3} , MSE
NAS-GRU	early stopping	patience 10, max 50 epochs, validation fraction 0.1
NAS-GRU	lookback / batch size	24 h / 256
NAS-GRU	seeds	{42, 43, 44, 45, 46}; per-city seed-mean before any test
Chronos-Bolt (zero-shot)	checkpoint	amazon/chronos-bolt-small (47,718,016 parameters), no training or fine-tuning
Chronos-Bolt (zero-shot)	context / point forecast	672 h (four weeks) / predicted mean
Chronos-Bolt + covariates	covariate model	ridge regression ($\alpha = 1.0$) on standardized calendar + weather features
Chronos-Bolt + covariates	scheme	FM forecasts the residual of the ridge fit; ridge horizon prediction added back

Table S3: Split-conformal 95% interval calibration, pooled per data tier. Empirical coverage and mean interval width (target units) pooled across cities within each tier, under the deployable causal-covariate configuration and the perfect-foresight upper bound. Covariate-free tiers (seasonal-naïve, NAS-GRU, zero-shot Chronos-Bolt) are identical in both configurations by construction. **(a)** PM2.5. **(b)** Temperature. Per-city calibration in Tables S5 and S6.

(a) PM2.5

Data tier	Model	Coverage (causal)	Width (causal)	Coverage (perfect)	Width (perfect)
rich	Chronos-Bolt (zero-shot)	0.929	18.47	0.929	18.47
rich	Chronos-Bolt + covariates	0.910	17.80	0.936	19.48
rich	LightGBM (specialist)	0.935	21.05	0.926	17.69
rich	NAS-GRU	0.914	18.78	0.914	18.78
rich	Seasonal-naïve	0.970	35.07	0.970	35.07
scarce	Chronos-Bolt (zero-shot)	0.931	108.64	0.931	108.64
scarce	Chronos-Bolt + covariates	0.928	123.94	0.931	112.69
scarce	LightGBM (specialist)	0.913	105.83	0.916	102.17
scarce	NAS-GRU	0.931	107.96	0.931	107.96
scarce	Seasonal-naïve	0.949	163.85	0.949	163.85

(b) Temperature

Data tier	Model	Coverage (causal)	Width (causal)	Coverage (perfect)	Width (perfect)
rich	Chronos-Bolt (zero-shot)	0.900	10.72	0.900	10.72
rich	Chronos-Bolt + covariates	0.923	23.91	0.892	9.17
rich	LightGBM (specialist)	0.923	9.56	0.958	7.70
rich	NAS-GRU	0.910	11.75	0.910	11.75
rich	Seasonal-naïve	0.934	18.04	0.934	18.04
scarce	Chronos-Bolt (zero-shot)	0.936	5.90	0.936	5.90
scarce	Chronos-Bolt + covariates	0.939	13.79	0.943	3.79
scarce	LightGBM (specialist)	0.944	4.38	0.949	2.68
scarce	NAS-GRU	0.950	6.02	0.950	6.02
scarce	Seasonal-naïve	0.968	8.96	0.968	8.96

Table S4: Full Diebold–Mariano pairwise win matrices (perfect-foresight panel). Per-pair significant-win counts (raw and FDR-adjusted) across the 29 cities. (a) PM2.5. (b) Temperature.

(a) PM2.5

Model pair	Cities	Sig. (raw)	A wins (raw)	B wins (raw)	A wins (FDR)	B wins (FDR)	Median P
Chronos-Bolt (zero-shot) vs Chronos-Bolt + covariates	29	5	4	1	2	1	0.295
Chronos-Bolt (zero-shot) vs NAS-GRU	29	6	5	1	2	0	0.308
Chronos-Bolt + covariates vs NAS-GRU	29	6	3	3	2	1	0.270
LightGBM (specialist) vs Chronos-Bolt (zero-shot)	29	3	2	1	0	0	0.429
LightGBM (specialist) vs Chronos-Bolt + covariates	29	3	3	0	2	0	0.486
LightGBM (specialist) vs NAS-GRU	29	7	6	1	6	1	0.384
Seasonal-naïve vs Chronos-Bolt (zero-shot)	29	18	0	18	0	13	0.029
Seasonal-naïve vs Chronos-Bolt + covariates	29	12	0	12	0	9	0.080
Seasonal-naïve vs LightGBM (specialist)	29	13	0	13	0	8	0.056
Seasonal-naïve vs NAS-GRU	29	15	0	15	0	11	0.047

(b) Temperature

Model pair	Cities	Sig. (raw)	A wins (raw)	B wins (raw)	A wins (FDR)	B wins (FDR)	Median P
Chronos-Bolt (zero-shot) vs Chronos-Bolt + covariates	29	5	2	3	2	1	0.370
Chronos-Bolt (zero-shot) vs NAS-GRU	29	9	8	1	6	0	0.107
Chronos-Bolt + covariates vs NAS-GRU	29	14	12	2	12	2	0.051
LightGBM (specialist) vs Chronos-Bolt (zero-shot)	29	11	11	0	6	0	0.167
LightGBM (specialist) vs Chronos-Bolt + covariates	29	9	9	0	6	0	0.153
LightGBM (specialist) vs NAS-GRU	29	20	20	0	20	0	0.003
Seasonal-naïve vs Chronos-Bolt (zero-shot)	29	20	0	20	0	16	0.017
Seasonal-naïve vs Chronos-Bolt + covariates	29	21	0	21	0	20	0.010
Seasonal-naïve vs LightGBM (specialist)	29	26	0	26	0	26	0.001
Seasonal-naïve vs NAS-GRU	29	16	2	14	2	13	0.022

Table S5: Per-city split-conformal calibration, PM2.5 (causal-covariate configuration). Empirical coverage (nominal 0.95) and mean interval width ($\mu\text{g m}^{-3}$) per city and tier.

City	Tier	Chronos cov.	Chronos width	Chronos+cov cov.	Chronos+cov width	LightGBM cov.	LightGBM width	NAS-GRU cov.	NAS-GRU width	Naïve cov.	Naïve width
Accra	scarce	0.86	24.5	0.89	32.8	0.86	25.4	0.90	29.2	0.93	52.0
Addis Ababa	scarce	0.92	57.7	0.93	68.5	0.81	45.7	0.89	59.2	0.94	84.0
Amsterdam	rich	0.82	5.3	0.92	6.8	0.92	7.9	1.00	12.4	1.00	34.8
Bangkok	rich	0.86	4.6	1.00	18.6	0.97	5.9	0.90	11.1	0.72	17.0
Berlin	rich	1.00	10.4	0.93	8.3	0.93	7.4	0.82	14.6	0.99	14.8
Bogota	scarce	1.00	47.2	0.96	41.6	1.00	43.0	1.00	41.4	1.00	54.0
Delhi	scarce	0.92	339.2	0.92	427.6	0.92	354.2	0.96	454.9	0.89	462.0
Dhaka	scarce	0.99	327.5	0.97	306.1	0.97	284.9	0.97	298.4	0.99	422.0
Hanoi	scarce	0.97	100.5	0.99	81.6	1.00	99.8	0.94	85.2	0.92	116.0
Jakarta	scarce	1.00	85.1	1.00	86.5	0.93	55.8	0.97	55.8	0.90	74.0
Kampala	scarce	0.97	125.1	0.94	106.6	0.90	85.2	0.88	86.4	0.90	158.0
Karachi	scarce	0.89	28.6	0.94	36.0	0.81	26.5	0.93	32.1	0.82	34.0
Krakow	rich	0.60	40.3	0.60	36.5	0.82	37.2	0.89	42.6	0.94	113.2
Lagos	scarce	0.79	22.4	0.83	31.1	0.79	20.7	0.79	20.0	0.97	42.2
Lahore	scarce	0.90	216.5	0.78	346.6	0.94	267.9	0.86	204.1	1.00	576.0
Lima	scarce	0.85	34.7	0.90	37.1	0.86	36.0	0.96	46.0	1.00	70.8
Los Angeles	rich	1.00	23.3	1.00	26.1	1.00	40.4	1.00	24.0	1.00	28.5
Madrid	rich	1.00	25.1	1.00	16.1	0.99	11.2	0.97	10.6	1.00	28.0
Manila	scarce	0.92	47.5	0.89	48.6	0.94	64.0	0.92	59.1	0.97	100.8
Mexico City	rich	0.86	27.7	0.81	24.9	1.00	61.7	1.00	44.9	1.00	64.0
Mumbai	scarce	1.00	143.2	1.00	187.0	0.96	142.3	1.00	116.2	1.00	160.0
Nairobi	scarce	1.00	29.9	0.97	21.5	1.00	36.2	0.99	31.3	1.00	52.0
New York	rich	0.96	12.3	0.99	12.4	0.89	11.9	0.81	11.5	1.00	21.4
Paris	rich	1.00	38.9	0.69	28.0	0.96	37.4	0.88	24.9	1.00	47.2
Santiago	rich	1.00	26.4	0.94	27.2	1.00	28.2	1.00	27.1	1.00	30.0
Seoul	rich	0.96	14.8	0.92	15.3	0.97	18.3	0.83	14.4	1.00	38.0
Sydney	rich	0.97	9.8	1.00	9.7	0.81	12.6	0.96	10.6	0.93	13.8
Toronto	rich	1.00	15.2	0.94	12.8	0.86	9.6	0.81	9.6	1.00	18.0
Vienna	rich	0.97	4.5	1.00	6.5	0.97	5.1	0.93	4.4	1.00	22.4

Table S6: Per-city split-conformal calibration, temperature (causal-covariate configuration). Empirical coverage (nominal 0.95) and mean interval width (°C) per city and tier.

City	Tier	Chronos cov.	Chronos width	Chronos+cov cov.	Chronos+cov width	LightGBM cov.	LightGBM width	NAS-GRU cov.	NAS-GRU width	Naïve cov.	Naïve width
Accra	scarce	0.88	1.8	0.92	10.0	0.88	2.3	0.82	2.7	1.00	4.6
Addis Ababa	scarce	1.00	13.2	1.00	21.2	0.94	4.4	0.96	6.2	0.97	17.2
Amsterdam	rich	1.00	13.7	1.00	30.1	1.00	15.0	1.00	14.0	1.00	16.0
Bangkok	rich	0.99	4.9	0.99	12.9	0.99	5.2	0.99	4.8	0.97	12.8
Berlin	rich	0.78	11.1	0.97	32.3	0.81	13.7	0.89	17.1	0.94	27.0
Bogota	scarce	0.78	3.6	0.81	18.2	0.86	3.5	0.86	5.7	0.89	5.6
Delhi	scarce	0.81	9.7	0.94	18.0	0.86	6.0	0.97	14.8	0.96	27.6
Dhaka	scarce	1.00	10.0	0.92	14.1	1.00	7.0	1.00	7.6	0.99	8.8
Hanoi	scarce	1.00	5.8	1.00	7.9	1.00	7.8	1.00	7.8	0.97	4.8
Jakarta	scarce	1.00	3.0	0.93	13.2	0.99	2.9	1.00	4.0	0.90	2.8
Kampala	scarce	1.00	16.2	1.00	16.4	1.00	7.0	1.00	11.8	1.00	20.8
Karachi	scarce	0.83	1.3	0.97	12.0	0.85	1.6	0.96	3.3	1.00	2.4
Krakow	rich	0.79	4.3	0.78	5.8	0.82	5.3	0.54	3.3	1.00	25.8
Lagos	scarce	0.82	2.1	0.83	5.7	0.83	2.6	0.89	2.7	1.00	6.0
Lahore	scarce	1.00	7.3	0.97	17.5	1.00	7.3	1.00	8.3	1.00	8.4
Lima	scarce	0.99	2.1	0.96	9.7	1.00	2.0	0.90	2.1	1.00	7.6
Los Angeles	rich	1.00	33.2	0.94	29.9	0.97	10.1	1.00	17.7	0.88	26.6
Madrid	rich	0.86	7.3	0.74	12.8	0.79	8.2	0.99	7.9	0.79	10.2
Manila	scarce	1.00	3.5	0.94	11.0	1.00	3.2	0.99	3.6	0.94	4.6
Mexico City	rich	0.58	4.1	0.85	20.5	0.72	4.4	0.58	3.5	0.79	13.2
Mumbai	scarce	0.94	4.7	0.89	12.5	0.97	4.4	0.92	5.9	0.89	7.0
Nairobi	scarce	1.00	4.1	1.00	19.6	0.99	3.8	0.99	3.7	1.00	6.2
New York	rich	0.96	8.4	0.96	31.9	0.94	6.9	0.93	5.1	1.00	18.2
Paris	rich	1.00	9.0	1.00	23.2	1.00	13.6	1.00	15.3	1.00	15.0
Santiago	rich	0.92	12.9	0.92	29.0	0.93	8.7	1.00	16.1	1.00	23.6
Seoul	rich	1.00	10.0	1.00	29.1	1.00	10.9	1.00	10.5	1.00	19.8
Sydney	rich	1.00	14.4	0.88	11.5	1.00	12.9	1.00	20.1	1.00	18.2
Toronto	rich	0.78	10.5	1.00	42.2	1.00	11.1	0.93	13.8	0.96	16.6
Vienna	rich	0.94	6.4	0.90	23.8	0.94	8.0	0.89	15.1	0.74	9.6

Table S7: Measured-energy repeatability. codecarbon-measured joules per 1,000 forecasts, five repetitions on three cities per tier, with the coefficient of variation and the 20% gate flag; flagged cells are reported as ranges in Table 3.

City	Tier	Reps	Mean (J/1k)	SD (J/1k)	SD/mean	>20% gate
Beijing	Chronos-Bolt (zero-shot)	5	1,229	240.9	0.196	False
Beijing	Chronos-Bolt + covariates	5	763	102.3	0.134	False
Beijing	LightGBM (specialist)	5	15,061	2,373.7	0.158	False
Beijing	NAS-GRU	5	13,237	2,690.1	0.203	True
Beijing	Seasonal-naïve	5	2	0.4	0.200	False
Nairobi	Chronos-Bolt (zero-shot)	5	1,946	1,940.3	0.997	True
Nairobi	Chronos-Bolt + covariates	5	842	108.9	0.129	False
Nairobi	LightGBM (specialist)	5	9,461	835.8	0.088	False
Nairobi	NAS-GRU	5	7,298	104.6	0.014	False
Nairobi	Seasonal-naïve	5	16	19.3	1.203	True
Seoul	Chronos-Bolt (zero-shot)	5	1,025	141.3	0.138	False
Seoul	Chronos-Bolt + covariates	5	781	94.2	0.120	False
Seoul	LightGBM (specialist)	5	10,832	406.1	0.037	False
Seoul	NAS-GRU	5	7,496	440.4	0.059	False
Seoul	Seasonal-naïve	5	9	15.2	1.684	True

Table S9: Horizon-48 panel accuracy. MASE (mean $\pm$ sd across cities) per tier at the 48-hour secondary horizon, perfect-foresight covariates, split by data tier (rich, $n = 14$ cities; scarce, $n = 15$). **(a)** PM2.5. **(b)** Temperature. The specialist’s temperature lead over zero-shot Chronos-Bolt widens with horizon exactly as the perfect-foresight-artifact hypothesis predicts: from 0.259 MASE pooled at $h = 24$ to 0.370 (rich) and 0.425 (scarce) at $h = 48$, since a perfect weather forecast is worth more the further ahead the model must look.

(a) PM2.5		
Tier	rich	scarce
Chronos-Bolt (zero-shot)	0.547 ± 0.208	0.892 ± 0.385
Chronos-Bolt + covariates	0.642 ± 0.245	0.902 ± 0.368
LightGBM (specialist)	0.535 ± 0.173	0.885 ± 0.407
NAS-GRU	0.639 ± 0.262	0.988 ± 0.435
Seasonal-naïve	0.843 ± 0.432	1.242 ± 0.545

(b) Temperature		
Tier	rich	scarce
Chronos-Bolt (zero-shot)	1.094 ± 0.315	0.958 ± 0.398
Chronos-Bolt + covariates	0.982 ± 0.351	0.710 ± 0.260
LightGBM (specialist)	0.724 ± 0.316	0.533 ± 0.278
NAS-GRU	1.345 ± 0.551	1.238 ± 0.649
Seasonal-naïve	1.926 ± 0.749	1.736 ± 0.972

Table S10: E4 transfer-vs-zero-shot full grid. Per-strategy mean MASE ($\pm$ sd) at each nominal fine-tune fraction, with the mean actual training hours used (nominal fractions floor differently per strategy, so actual hours are reported alongside).

Strategy	Nominal fraction (%)	MASE (mean $\pm$ sd)	Actual training hours (mean)	Cities
Chronos-Bolt (zero-shot)	—	0.843 ± 0.354	0	15
LightGBM (refit on budget)	1	0.941 ± 0.311	202	15
LightGBM (refit on budget)	10	0.944 ± 0.383	599	15
LightGBM (refit on budget)	100	0.858 ± 0.382	5,995	15
NAS-GRU (transfer + fine-tune)	0	0.899 ± 0.340	0	15
NAS-GRU (transfer + fine-tune)	1	0.915 ± 0.331	66	15
NAS-GRU (transfer + fine-tune)	10	0.888 ± 0.340	599	15
NAS-GRU (transfer + fine-tune)	100	0.876 ± 0.374	5,995	15

Table S12: Cost-assumption sensitivity of the decision maps (causal configuration, matching main-text Fig. 5). Winner-cell flip rate for each regime run under an electricity-price $\times$ PUE grid (price and PUE act as linear rescalings of the cost-penalty coefficient λ ; the central assumption 0.15 USD/kWh $\times$ PUE 1.4 has multiplier 1 by definition). No cells flip at the central assumption; at most 40% flip at the most extreme corners of the grid.

Regime run	Price (\$/kWh)	PUE	Effective λ multiplier	Cells	Flipped	Flip rate
Beijing PM2.5	0.05	1.0	0.238	25	4	0.16
Beijing PM2.5	0.05	1.4	0.333	25	4	0.16
Beijing PM2.5	0.05	2.0	0.476	25	3	0.12
Beijing PM2.5	0.15	1.0	0.714	25	3	0.12
Beijing PM2.5	0.15	1.4	1.000	25	0	0.00
Beijing PM2.5	0.15	2.0	1.429	25	0	0.00
Beijing PM2.5	0.30	1.0	1.429	25	0	0.00
Beijing PM2.5	0.30	1.4	2.000	25	2	0.08
Beijing PM2.5	0.30	2.0	2.857	25	3	0.12
Beijing Temperature	0.05	1.0	0.238	25	4	0.16
Beijing Temperature	0.05	1.4	0.333	25	3	0.12
Beijing Temperature	0.05	2.0	0.476	25	2	0.08
Beijing Temperature	0.15	1.0	0.714	25	1	0.04
Beijing Temperature	0.15	1.4	1.000	25	0	0.00
Beijing Temperature	0.15	2.0	1.429	25	1	0.04
Beijing Temperature	0.30	1.0	1.429	25	1	0.04
Beijing Temperature	0.30	1.4	2.000	25	2	0.08
Beijing Temperature	0.30	2.0	2.857	25	3	0.12
Nairobi PM2.5	0.05	1.0	0.238	20	7	0.35
Nairobi PM2.5	0.05	1.4	0.333	20	5	0.25
Nairobi PM2.5	0.05	2.0	0.476	20	4	0.20
Nairobi PM2.5	0.15	1.0	0.714	20	3	0.15
Nairobi PM2.5	0.15	1.4	1.000	20	0	0.00
Nairobi PM2.5	0.15	2.0	1.429	20	2	0.10
Nairobi PM2.5	0.30	1.0	1.429	20	2	0.10
Nairobi PM2.5	0.30	1.4	2.000	20	8	0.40
Nairobi PM2.5	0.30	2.0	2.857	20	8	0.40
Nairobi Temperature	0.05	1.0	0.238	20	4	0.20
Nairobi Temperature	0.05	1.4	0.333	20	4	0.20
Nairobi Temperature	0.05	2.0	0.476	20	2	0.10
Nairobi Temperature	0.15	1.0	0.714	20	0	0.00
Nairobi Temperature	0.15	1.4	1.000	20	0	0.00
Nairobi Temperature	0.15	2.0	1.429	20	4	0.20
Nairobi Temperature	0.30	1.0	1.429	20	4	0.20
Nairobi Temperature	0.30	1.4	2.000	20	7	0.35
Nairobi Temperature	0.30	2.0	2.857	20	8	0.40
Seoul PM2.5	0.05	1.0	0.238	25	5	0.20
Seoul PM2.5	0.05	1.4	0.333	25	4	0.16
Seoul PM2.5	0.05	2.0	0.476	25	4	0.16
Seoul PM2.5	0.15	1.0	0.714	25	0	0.00
Seoul PM2.5	0.15	1.4	1.000	25	0	0.00
Seoul PM2.5	0.15	2.0	1.429	25	1	0.04
Seoul PM2.5	0.30	1.0	1.429	25	1	0.04
Seoul PM2.5	0.30	1.4	2.000	25	9	0.36
Seoul PM2.5	0.30	2.0	2.857	25	10	0.40
Seoul Temperature	0.05	1.0	0.238	25	0	0.00
Seoul Temperature	0.05	1.4	0.333	25	0	0.00
Seoul Temperature	0.05	2.0	0.476	25	0	0.00
Seoul Temperature	0.15	1.0	0.714	25	0	0.00
Seoul Temperature	0.15	1.4	1.000	25	0	0.00
Seoul Temperature	0.15	2.0	1.429	25	0	0.00

Table S13: Energy train/inference decomposition and amortization crossover. One-time training energy per fit versus per-forecast inference energy for the specialist and the zero-shot foundation model, on CPU and GPU, and the crossover count (forecasts a once-trained specialist serves before its amortized energy per forecast falls to the foundation model’s inference energy). codecarbon measurements at sub-kilojoule scale are noisy (the GPU column is less stable; one negative inference estimate is a measurement artifact) — the CPU rows are the reliable signal.

City	Device	Train energy / fit (J)	LightGBM infer (J/forecast)	Chronos infer (J/forecast)	Crossover (forecasts)
Beijing	CPU	317	0.033	0.112	4,041
Beijing	GPU	550	-0.243	1.942	252
Nairobi	CPU	203	0.065	0.111	4,416
Nairobi	GPU	348	0.075	0.322	1,411
Seoul	CPU	246	0.022	0.110	2,778
Seoul	GPU	423	0.130	0.327	2,147

Table S14: Perfect-foresight panel accuracy (upper bound). Table 1 recomputed with perfect-foresight covariates for every covariate-using tier — an upper bound on the specialist’s reach that assumes a perfect forecast of its own inputs. The specialist reaches 0.533 on temperature with 6 FDR-significant wins here; under deployable causal covariates (Table 1) it has none.

Tier	Domain	MASE (mean $\pm$ sd)	MAE (mean $\pm$ sd)	RMSE (mean $\pm$ sd)	Avg rank	FDR wins (tier/FM)
Chronos-Bolt (zero-shot)	PM2.5	0.662 $\pm$ 0.368	10.553 $\pm$ 13.524	15.093 $\pm$ 19.273	1.83	—
Chronos-Bolt + covariates	PM2.5	0.730 $\pm$ 0.376	11.745 $\pm$ 15.266	15.959 $\pm$ 20.478	3.14	1 / 2
LightGBM (specialist)	PM2.5	0.662 $\pm$ 0.371	10.845 $\pm$ 14.060	14.810 $\pm$ 18.895	2.17	0 / 0
NAS-GRU	PM2.5	0.734 $\pm$ 0.352	11.426 $\pm$ 14.931	15.592 $\pm$ 20.791	3.07	0 / 2
Seasonal-naïve	PM2.5	1.026 $\pm$ 0.530	16.505 $\pm$ 21.235	22.274 $\pm$ 28.983	4.79	0 / 13
Chronos-Bolt (zero-shot)	Temp.	0.792 $\pm$ 0.282	1.303 $\pm$ 0.717	1.819 $\pm$ 1.040	2.90	—
Chronos-Bolt + covariates	Temp.	0.718 $\pm$ 0.278	1.228 $\pm$ 0.795	1.559 $\pm$ 1.012	2.41	1 / 2
LightGBM (specialist)	Temp.	0.533 $\pm$ 0.208	0.888 $\pm$ 0.550	1.153 $\pm$ 0.730	1.28	6 / 0
NAS-GRU	Temp.	0.999 $\pm$ 0.481	1.626 $\pm$ 1.093	2.070 $\pm$ 1.261	3.59	0 / 6
Seasonal-naïve	Temp.	1.686 $\pm$ 1.050	2.716 $\pm$ 1.780	3.300 $\pm$ 2.085	4.83	0 / 16

Table S15: Post-cutoff (unseen-data) check. Zero-shot Chronos-Bolt versus the LightGBM specialist evaluated only on strictly post-2024-10 PM2.5 windows, which postdate the model’s pretraining corpus, for the 10 cities with sufficient recent coverage. The foundation model remains competitive (mean MASE 0.415 vs 0.395; better in 6 of 10 cities), so the PM2.5 result is not a pretraining-exposure artifact.

City	Post-cutoff hours	Chronos MASE	LightGBM MASE	Naive MASE
Amsterdam	11,228	0.243	0.249	0.814
Bangkok	4,823	0.133	0.196	0.428
Berlin	15,276	0.238	0.255	0.493
Lahore	3,369	0.468	0.512	0.734
Los Angeles	7,757	0.532	0.521	0.710
Madrid	6,431	0.690	0.346	0.755
Manila	10,992	0.935	0.911	1.655
New York	8,499	0.425	0.470	0.631
Seoul	15,237	0.321	0.339	0.742
Vienna	6,241	0.167	0.151	0.528

Table S16: Sensitivity of the TOST equivalence verdict to the margin. For each of the seven pre-specified tie comparisons: the paired point difference (positive favours the zero-shot foundation model), the TOST 90% confidence interval, the smallest equivalence margin at which the interval fits inside $\pm\delta$ (“min. margin”), and the verdict ($\checkmark$ = equivalent) at a strict ($\delta = 0.02$), the main-text ($\delta = 0.05$), and a lenient ($\delta = 0.10$) MASE margin. The main-text 0.05 margin was fixed before the test was run but is not a registered pre-campaign quantity (Methods); the table makes the margin dependence explicit. At $\delta = 0.05$ only the perfect-foresight PM2.5 comparison is equivalent; a lenient 0.10 margin additionally admits causal PM2.5 and the 100%-budget E4 comparison.

Comparison	Point diff	TOST 90% CI	Min. margin	$\delta=0.02$	$\delta=0.05$	$\delta=0.10$
PM2.5, perfect-foresight (specialist vs FM)	-0.000	[-0.042, +0.042]	0.042	–	$\checkmark$	$\checkmark$
PM2.5, causal (specialist vs FM)	+0.030	[-0.011, +0.071]	0.071	–	–	$\checkmark$
Temperature, causal (specialist vs FM)	-0.048	[-0.114, +0.018]	0.114	–	–	–
E4 transfer 0\{\}\% vs zero-shot	+0.056	[+0.005, +0.107]	0.107	–	–	–
E4 transfer 1\{\}\% vs zero-shot	+0.072	[+0.015, +0.129]	0.129	–	–	–
E4 transfer 10\{\}\% vs zero-shot	+0.045	[-0.012, +0.102]	0.102	–	–	–
E4 transfer 100\{\}\% vs zero-shot	+0.033	[-0.009, +0.075]	0.075	–	–	$\checkmark$

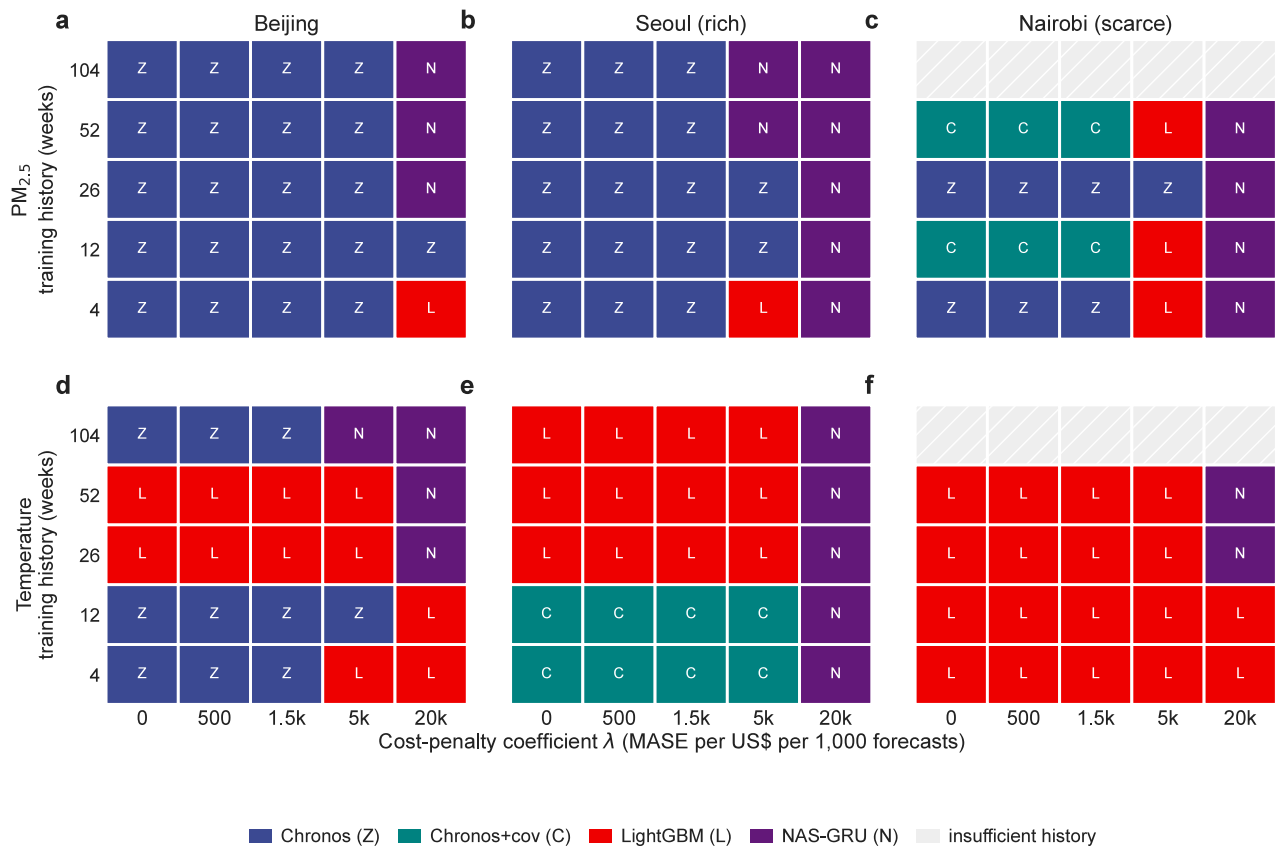


Figure S1: Cost-adjusted deployment winner maps, perfect-foresight covariates (upper bound). As Figure 5 of the main text but with perfect-foresight covariates for every covariate-using tier (panels **a–f** as in Fig. 6; cell letters duplicate the colour coding for grayscale readers). The temperature maps favour the specialist here (Beijing 11/25, Seoul 12/25, Nairobi 18/20 cells); under the deployable causal covariates of the main Figure 6 that dominance collapses.

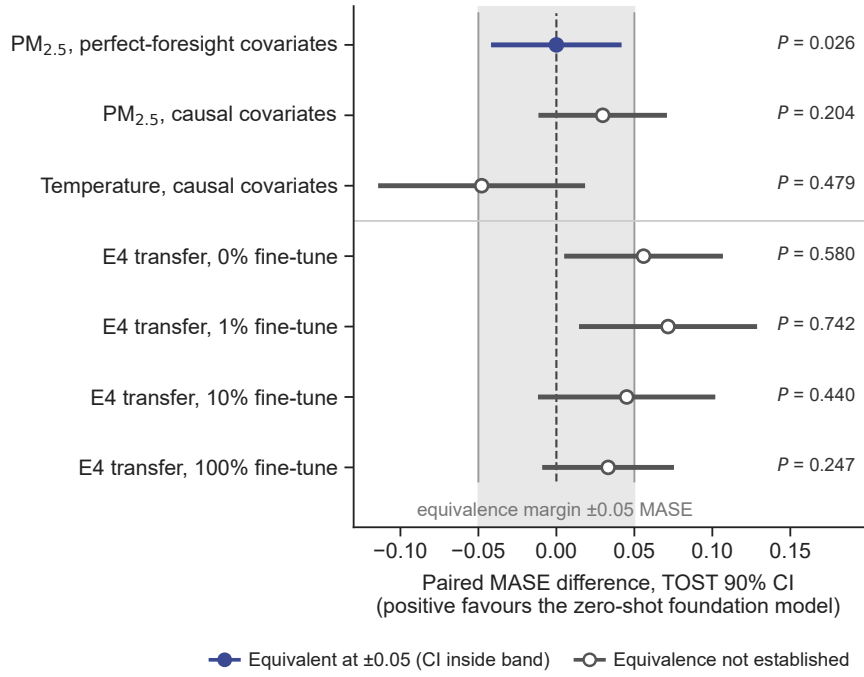


Figure S2: TOST equivalence tests for all seven pre-specified tie comparisons. Paired MASE difference (positive favours the zero-shot foundation model) with the TOST 90% confidence interval against the pre-specified ± 0.05 -MASE equivalence margin (shaded band); P values are the paired TOST P at that margin. Equivalence is established only where the whole interval lies inside the band (filled marker): the perfect-foresight PM_{2.5} specialist-versus-foundation-model comparison ($n = 29$ cities; E4 rows $n = 15$ scarce cities). All other comparisons — causal PM_{2.5}, causal temperature, and every E4 fine-tune budget — remain “no significant difference” verdicts, not demonstrated equivalence.